

EXPERIMENT NO. 19:

Implementation of Naive Bayes Classification for Bank Loan Prediction

Aim:

To implement the Naive Bayes Classification Algorithm in Python for predicting whether a bank loan will be Approved or Rejected based on customer details.

Procedure:

1. Import the required Python libraries.
2. Create a sample bank loan dataset.
3. Encode the categorical attributes into numerical values.
4. Separate the input features and target class.
5. Split the dataset into training and testing sets.
6. Create and train a Gaussian Naive Bayes classifier.
7. Predict loan approval for the test data.
8. Calculate and display the accuracy.
9. Predict the loan status of a new customer.

Program:

```
# Naive Bayes Classification for Bank Loan Prediction

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import accuracy_score

# Sample Bank Loan Dataset

data = {

    'Age': [25, 35, 45, 30, 50, 28, 40, 55, 32, 48, 26, 42],
```

```

'Income': [25000, 45000, 60000, 35000, 80000, 30000, 55000, 90000, 40000, 70000,
28000, 65000],

'Loan_Amount': [100000, 200000, 300000, 150000, 400000, 120000, 250000, 500000,
180000, 350000, 100000, 300000],

'Credit_Score': [600, 700, 750, 650, 800, 620, 720, 820, 680, 780, 610, 740],

'Loan_Status': ['Rejected', 'Approved', 'Approved', 'Rejected', 'Approved', 'Rejected',
'Approved', 'Approved', 'Rejected', 'Approved', 'Rejected', 'Approved']

}

df = pd.DataFrame(data)

# Input features and target

X = df[['Age', 'Income', 'Loan_Amount', 'Credit_Score']]

y = df['Loan_Status']

# Encode target values

encoder = LabelEncoder()

y = encoder.fit_transform(y)

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=42)

# Create Naive Bayes model

model = GaussianNB()

# Train the model

model.fit(X_train, y_train)

# Predict test data

y_pred = model.predict(X_test)

# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)

```

```

print("Actual Loan Status:")

print(encoder.inverse_transform(y_test))

print("\nPredicted Loan Status:")

print(encoder.inverse_transform(y_pred))

print("\nAccuracy:")

print(round(accuracy * 100, 2), "%")

# Predict a new customer

new_customer = [[38, 55000, 250000, 720]]

prediction = model.predict(new_customer)

print("\nNew Customer Loan Status:")

print(encoder.inverse_transform(prediction)[0])

```

Output:

```

*** Actual Loan Status:
['Rejected' 'Approved' 'Rejected']

Predicted Loan Status:
['Rejected' 'Approved' 'Rejected']

Accuracy:
100.0 %

New Customer Loan Status:
Approved
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but GaussianNB was fitted with feature names
warnings.warn(

```

Result:

The Naive Bayes Classification Algorithm was successfully implemented for Bank Loan Prediction. The model predicted whether a loan would be approved or rejected based on the customer's age, income, loan amount, and credit score, and successfully classified a new customer's loan as Approved.